

Chapter 1

Creating Virtuous Cycles for DNA Barcoding: A Case Study in Science Innovation, Entrepreneurship, and Diplomacy

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Abstract

This chapter on the history of the DNA barcoding enterprise attempts to set the stage for the more scholarly contributions in this volume by addressing the following questions. How did the DNA barcoding enterprise begin? What were its goals, how did it develop, and to what degree are its goals being realized? We have taken a keen interest in the barcoding movement and its relationship to taxonomy, collections, and biodiversity informatics more broadly considered. This chapter integrates our two different perspectives on barcoding. DES was the Executive Secretary of the Consortium for the Barcode of Life from 2004 to 2017, with the mission to support the success of DNA barcoding without being directly involved in generating barcode data. RDMP viewed barcoding as an important entry into the landscape of biodiversity data, with many potential linkages to other components of that landscape. We also saw it as a critical step toward the era of international genomic research that was sure to follow. Like the Mercury Program that paved the way for lunar landings by the Apollo Program, we saw DNA barcoding as the proving grounds for the interdisciplinary and international cooperation that would be needed for success of whole-genome research.

Key words Barcoding, Informatics, "Data curation", Incentives, "Historical trends"

1 The Founding Events of DNA Barcoding, 2003–2005

The DNA Barcoding initiative, like other innovative scientific enterprises such as the Human Genome Project, can be viewed through the prism of events, both large and small. Why do the events in a few scientific endeavors set off chain reactions that result in stunning new insights and open new, exciting areas of research? What is it about highly successful enterprises that enable them to overcome obstacles like doubt, skepticism, and competing interests? What can we learn from the sequence of events in the early history of DNA barcoding that can inform the way biodiversity research is conducted in the future?

DNA barcoding began with two publication events: Folmer et al. (1994 [1]) suggested "universal" PCR primers that were effective for the mitochondrial cytochrome c oxidase subunit I

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gene (COI) across 11 invertebrate phyla and Hebert et al. (2003 [2]) extended the 1994 analysis of COI sequences to include 15 insect orders. They went on to note the potential parallel between the COI subregion and the Universal Product Code, noting "these sequences can be viewed as genetic 'barcodes' that are embedded in every cell."

Jesse Ausubel (Rockefeller University and Sloan Foundation Program Director) attended a Canadian marine biology workshop in early 2002, and he heard Hebert's presentation of DNA barcoding a year prior to publication. Ausubel saw the broad potential impact of the idea, and two planning workshops were held at the Banbury Conference Center in March and September 2003 with Sloan Foundation support. These workshops led, respectively, to publication of Stoeckle, 2003 [3], and a successful proposal to the Sloan Foundation to establish the Consortium for the Barcode of Life[1] (CBOL), hosted by the Smithsonian Institution's National Museum of Natural History starting in May 2004. CBOL's mission, developed during the Banbury workshops, was to promote the success of DNA barcoding by acting as a strategic partner and a complement to Hebert's lab at the University of Guelph and other leading research centers and biorepositories that would be at the forefront of barcoding efforts.

Less than two years after the publication of Hebert et al., 2003, the First International Barcode of Life Conference took place at The Natural History Museum London on February 7–9, 2005, organized by CBOL and attended by more than 300 participants from approximately 50 countries. CBOL also helped organize a two-day conference, "Library & Laboratory: The marriage of research, data and taxonomic literature," immediately before the 2005 London Conference. The "Lib-Lab" conference launched the process leading to creation of the Biodiversity Heritage Library, the leading initiative to digitize the biodiversity research literature.

The London Conference volume, published in October that year (Savoleinen et al., eds., 2005) [4], laid out the principal directions that a global barcoding enterprise would take in the years that followed. Several articles in the Conference volume pointed out challenges that DNA barcoding would face. The following four chapters [5–8] are examples. COI was not considered an effective barcode for land plants, and other candidate regions had different limitations (Chase et al., 2005) [5]. In some taxonomic groups that lack distinctive features, DNA barcode data could have disproportionate impact on species diagnoses and identifications relative to morphology (e.g., red macroalgae; Saunders 2005) [6]. Body sizes are so small in groups like meiobenthics that entire specimens were consumed by DNA extraction, making preservation of voucher

[1] See https://web.archive.org/web/20170101084850/http://www.barcodeoflife.org/

specimens for the barcode sequence impossible. Environmental mixtures could yield barcode sequences, and species could be identified using "reverse taxonomy" (Markmann and Tautz, 2005) [7], or "molecular operating taxonomic units" (MOTUs) could be recognized and used rather than formal species names (Blaxter et al., 2005) [8].

Each of these challenges involved, to varying degrees, concerns over how barcode data would be integrated with morphological and other classes of organismal trait data in diagnosing, describing, and identifying species. DeSalle, Egan, and Siddall (2005) [9] wrote "The main problem that needs to be addressed in any attempt to determine the boundary of a species. . .is to avoid circular or tautological reasoning." They described how the discrimination of species could begin with any class of traits (molecular, morphological, geographic), as long as new species concepts had corroborating evidence from other traits.

With the advantages of hindsight, the London conference foreshadowed three principal areas of effort:

A. "Supply-side" barcoding would be led by taxonomic specialists interested in adding a standardized layer of diagnostic genetic data that could be comparable across and beyond the taxa in which they were most interested. These specialists would be the primary contributors to the global, standardized "DNA Barcoding Reference Library" (DBRL), because they have the best access to well-studied voucher specimens and knowledge of the trait data used to identify them (morphology, ecology, geographic distribution, behavior, development). Progress in populating DBRL would take the form of (1) increasing numbers of high-quality records in DBRL; (2) significant numbers of records in DBRL representing diverse taxonomic groups; (3) contributions to DBRL from all regions of the world, especially those with the greatest and least well-known biodiversity; (4) taxonomic revisions that would increase understanding of biodiversity and improve species-level taxonomy by integrating a "common yardstick" for judging similarity and differences; and (5) taxonomic discovery and published documentation of new species.

B. "Demand-side" barcoding would be driven by more applied researchers, such as those in regulatory agencies and private industry, and would rely on DBRL to assign unidentified specimens to known species. Reliable identification of unknowns would require adequate representation of those taxonomic groups in DBRL, thereby creating incentives for mutual support by supply- and demand-side barcoders. Success would take the form of funding for supply-side barcoding of taxonomic groups of interest to demand-side users. Three oft-cited examples were agencies responsible for detecting and

excluding agricultural pests at ports of entry; prosecuting traffickers in endangered species; and monitoring environmental quality using biodiversity surveys.

C. Technology and infrastructure development would be critical for meeting the enormous scale of the challenge and achieving economies of scales that would enable rapid progress. One contribution to the London Conference proceedings (Hajibabei et al., 2005) [10] reviewed the logistical challenges (primarily those that supply-side barcoding would face) and envisioned large, centralized barcoding facilities, akin to industrial manufacturing plants. The authors went through the analytical process from DNA extraction to sequencing to data management, quality control, and public data release. They noted that the most active barcoding projects had created production-line protocols to increase the production of barcode records. These vanguard projects (many of which contributed articles to the London conference proceedings) concentrated on specific taxonomic groups and geographic areas and could not address some of the ambitious goals for taxonomic and geographic breadth.

The London Conference, and the associated meeting on digitizing the biodiversity literature, generated great enthusiasm among participants. Coordinating committees were established in many countries and global initiatives on certain taxonomic groups were launched soon after the London conference ended.

2 The Expansion Phase, 2006–2011

In the years following the 2005 London Conference, Hebert's laboratory and informatics operations at the University of Guelph expanded rapidly. Funding from several Canadian sources enabled them to create a sophisticated barcoding workflow devoted to supply-side barcoding. Their specific goal was to produce more than 100,000 DBRL records per year. Their Barcode of Life Data Systems (BOLD) was developed, tested, and put into operation as a specialized public workbench and repository of barcode records (Ratnasingham and Hebert, 2007) [11]. For a period of time, selected projects with large numbers of well-documented voucher specimens received free barcoding services at Guelph's facilities. Public release of these records demonstrated how the Guelph workflow was overcoming the challenges articulated in [10], as long as voucher specimens with intact DNA and the required metadata were available.

In June 2007, Hebert's lab gathered researchers from 25 countries to plan the next phase of international collaboration for building DBRL. National committees formed and became

components of the International Barcode of Life (iBOL) launched in 2008. New funding from Canadian sources led iBOL to activate "Barcode 500K" in October 2010, a campaign to generate the first DBRL records for 500,000 species. The research at the University of Guelph also opened new activities for public engagement through citizen science biodiversity surveys, and intense sampling and barcoding of all taxa in study sites in boreal Canada.

During this same period, CBOL ramped up its outreach activities to build participation. For example, CBOL promoted the launch of large barcoding projects and engaged with national and international organizations whose acceptance of barcoding was critical.[2] CBOL's outreach efforts included three more International Barcode Conferences[3] which helped to build its membership to 200 institutions, government agencies, and private companies in more than 50 countries, each committing to the public release of their barcode records within one year of completion. CBOL organized and sponsored planning workshops and launch conferences for large-scale barcoding project, especially for birds[4] (with large collections of frozen tissue linked to study skins) and fish (with special importance to food security and the detection of species substitution in the marketplace; Ward, Hanner, and Hebert, 2009) [12]. CBOL also helped to promote demand-side barcoding projects in areas like environmental biomonitoring using barcodes (e.g., Pfrender and others, 2010) [13] and adoption of barcoding as an international phytosanitary protocol for biosecurity.[5] CBOL Working Groups were formed and supported to determine the optimal barcode region(s) for plants (CBOL Plant Working Group. 2009) [14]; develop and test new data analysis tools; and establish a data standard for INSDC records considered fit for use in assigning samples to named species based on a barcode sequence.[6] INSDC records were tagged with the keyword "BARCODE" if they met the standard's requirement for a voucher specimen identifier, a recognized species name, latitude and longitude of the collecting site, sequence length and quality, and links to trace files.

[2] A full listing of CBOL-sponsored events is available at https://web.archive.org/web/20140929191855/http://www.barcodeoflife.org/content/events/past

[3] Taipei in 2007, Mexico City in 2009, and Adelaide in 2011.

[4] All Birds Barcoding Initiative, ABBI: https://phe.rockefeller.edu/barcode/docs/ABBI-barcoding%20life%20takes%20flight%20v1.1.pdf

[5] CBOL hosted a meeting of the FAO's Technical Diagnostics Protocol Committee, International Plant Protection Commission, at the Smithsonian in July 2010, at which barcoding protocols for identifying agricultural pests were reviewed.

[6] CBOL's Database Working Group met in May and October 2005 and proposed the BARCODE Data Standard on November 6, 2005, updated March 26, 2009; see https://web.archive.org/web/20170221170746/http://www.barcodeoflife.org/sites/default/files/DWG_data_standards-Final.pdf

Digitization of the specimen records in natural history museums, herbaria, and other biodiversity repositories began in the 1960s and was becoming a high priority for these institutions by the time of the London conference. CBOL saw an important role for DNA barcoding as a key variable in this rapidly developing informatics infrastructure. Specifically, CBOL opened discussions with key bioinformatics initiatives with the goal of linking these interrelated data:

- Published DNA barcode sequence records
- Their voucher specimen records in biorepositories
- Their taxonomic names as described in authority databases like the Catalog of Life
- Digitized publications in which the voucher specimens or barcode records were cited

Working with several other public sources of information about biorepositories[7] CBOL created the Global Registry of Biodiversity Collections (GRBio) in 2007 [15], later expanded to the Global Registry of Science Collections (GRSciColl),[8] with coverage of all disciplines with collections. Two important functions of the registry were providing unique repository identifiers that can be cited in and retrieved from publications and facilitating user access to collections.

CBOL's program of work also addressed issues of science diplomacy, especially those involving international treaties such as the Convention on Biological Diversity (CBD), International Trade in Endangered Species (CITES), and the International Plant Treaty. These and other UN Conventions involved international access to biological samples which was becoming increasingly contentious during this period.[9] CBOL was successful in getting the following language accepted as part of Article 8 of the Nagoya Protocol (Secretariat of the Convention on Biological Diversity, 2011) [16]:

> In the development and implementation of its access and benefit-sharing legislation or regulatory requirements, each Party shall:
>
> 1. Create conditions to promote and encourage research which contributes to the conservation and sustainable use of biological diversity, particularly in developing countries,

[7] GRBio integrated data from GenBank's database of institutions, Index Herbariorum (curated by the NY Botanical Garden), and Biodiversity Collections Index (BCI).

[8] GRSciColl (https://www.gbif.org/grscicoll) is now operated by the Global Biodiversity Information Facility in Copenhagen.

[9] CBOL organized workshops on international access to genetic resources which produced a 2008 workshop report on Access and Benefit Sharing

> including through simplified measures on access for non-commercial research purposes, taking into account the need to address a change of intent for such research.

Access to well-preserved tissue samples from poorly known taxonomic groups and diversity-rich developing countries was becoming a greater problem through this period. As predicted in Hajibabei et al. (2005) [10], better DNA extraction and repair techniques, as well as new short-read sequencing platforms, were gradually making barcoding possible using older, compromised DNA samples. Nevertheless, there were vast numbers of well-preserved frozen tissue samples with complete metadata in biorepositories that had not yet been barcoded with easier methods. Stoeckle and Winkler (2009) [17] reported that the 30 largest collections of frozen bird tissues had voucher specimens for 40–100% of their frozen tissue samples. Of the nearly 10,000 recognized species of birds, more than 70% are represented in frozen tissue collections.

CBOL encouraged natural history museums and research zoos with large repositories of cryopreserved tissues to digitize these collections through DNA barcoding. Zoos were universally resistant to the idea of making DNA barcode data public, and few museums were willing to undertake large-scale barcoding of their frozen tissue collections. CBOL partnered with the Ornithology Department in the Smithsonian's National Museum of Natural History for a demonstration project leading to a data release publication. The public release on GenBank of 2808 bird barcode records was announced in November 2011 at the Fourth International Barcode Conference in Adelaide, Australia. This data release (Schindel et al., 2011) [18], designed to serve the functions proposed by the Wellcome Trust ([19]; discussed in the final section of this article), added the first barcode records for 1147 species, increasing the number of bird species with public barcodes by 91%.

3 Inflection Points and New Directions, 2011–2013

Two new trends emerged around the time of the 2011 conference in Adelaide; some were anecdotal while others were documented in retrospective studies a few years later. The Adelaide conference was the first at which researchers were voicing concern about accessing identified voucher specimens with well-preserved frozen tissue samples, especially from high diversity developing countries. Second, it was becoming increasingly difficult to obtain funding for large-scale supply-side production of thoroughly documented and reliable barcode records. Access to frozen tissue samples from identified voucher specimens was decreasing, so field collecting of new vouchers became necessary. As a result, basic taxonomic

research was needed to identify them, and reviewers and funders began to view these activities as infrastructure-building, not question-driven research.

Published proposals for ways to solve two major challenges appeared in a short time. First, "environmental barcoding" (Hajibabaei et al., 2011) [20] and "metabarcoding" (Taberlet et al., 2012) [21] were proposed as ways to extract and sequence DNA from alcohol-preserved freshwater assemblages and interstitial fluids from soil samples, respectively. High-throughput next-generation sequencing would produce short reads that could be assembled and compared with voucher-based BARCODE records. For example, Stoeckle, Soboleva, and Charlop-Powers Z. (2017) [22] used metabarcoding protocols to identify fish species and estimate their abundances from water samples.

Second, the "Barcode Index Number" (BINs; Ratnasingham and Hebert, 2013) [23] was proposed as a system of unique identifiers for statistical clusters of barcode sequences, reminiscent of MOTUs (Blaxter et al., 2005) [8]. In many cases, well-documented BARCODE sequences from identified voucher specimens might cluster within the variations assigned to a BIN that also contained barcode records without voucher specimens. This suggested, but could not confirm, that the sources of the vouchered and unvouchered metabarcodes belong to the same species. Schindel and Miller (2010) [24] had argued that systems of provisional nomenclature such as BINs are an "on-ramp to taxonomic names." They stressed that accelerating movement along this on-ramp would require informatics infrastructure (like BOLD) and community collaboration in assembling relevant samples and data. This crowd-sourced data curation could produce DRBL records with confirmed identifications in the manner described in [9]. Without such a process, the distinction would become blurred between vouchered and validated barcode records, on the one hand, and unvouchered or unvalidated records whose species identifications were ***inferred from*** other, perhaps unvalidated, barcode records, on the other hand.

Environmental barcoding and metabarcoding studies, combined with BINs and MOTUs, provided ways to greatly accelerate biomonitoring and other studies of species distributions, with one important drawback. Many demand-based barcoding initiatives are tied to regulatory frameworks (e.g., food security, endangered species protection, control of the import of invasive species). Such frameworks usually have their own specific standards for species identification, often tied to specimens held in secured collections that act as standards for species identification. Species names and distribution patterns based on metabarcoding and BINs may not follow the procedures required or meet the data quality standards expected for regulatory or legal actions. CBOL discovered this distinction in many government agencies, international

organizations, and different countries. The difference between academic barcoding studies and the expectations of applied, demand-side users surfaced as a common challenge during this period (discussed below).

4 Taking Stock, 2013–Present

During this phase, review articles began appearing on the progress being made producing well-documented and authoritative barcode records—records that meet the expectations established at the 2005 London Conference. Kvist (2013) [25] noted the low coverage within many taxonomic groups and uneven species-level coverage across animal phyla. He found that barcode records exist for a high proportion of known species only in "charismatic" groups such as Craniata and very low-diversity groups. Groups with high species diversity had more than 10% coverage only if they were the focus of a barcoding initiative (e.g., arthropods, with 15% of species represented in INSDC and BOLD, combined).

Porter and Hajibabei (2018) [26] provided a valuable status report on barcode records in GenBank and BOLD, with emphasis on their relevance to demand-side communities (e.g., freshwater biomonitoring, species protection). They found 2,530,418 COI eukaryote records in INSDC (3,587,391 as of July 1, 2023) and they echoed Kvist's 2013 observations of uneven taxonomic coverage, emphasizing the lack of barcode data for endangered species [23]. They noted that while well-studied groups have substantial barcode data, they are stored separately in GenBank and BOLD. They expressed concern at the lack of a system of regular synchronization between the two databases and reported inconsistent cross-referencing of records found in both. One important distinction is the proportion of BARCODE records in each. As of this writing, 1,145,016 COI eukaryote records in GenBank had the keyword BARCODE, representing 32% of all eukaryote COI records (3,587,391). In 2018, [24] reported that 28% of COI records in GenBank had the keyword BARCODE, while only 13–15% of the records in BOLD did. During the period 2010–2014, the number of new barcode records with recognized taxonomic names increased 46% per year, while records without them increased 66% per year (see below). In 2018, BOLD had 2,727,834 public COI records, only 46% of which were also represented in INSDC; and of these, approximately 25% had the BARCODE keyword and about 75% did not.

Page (2016) [27] presented an analysis of progress in the barcoding initiative as viewed from the informatics perspective. He noted that while the rate at which new species names are being published was more or less stable between 1995 and 2009, or growing slowly when averaged across groups, the public release

of barcode records has been accelerating. For example, he reported 531,469 GenBank records had been given the BARCODE keyword in between 2007 and 2016 (1,145,209 as of this writing), and the iBOL BioProject (PRJNA37833) had 194,727 records (338,079 now). However, fewer than half of new barcode records for invertebrates released on GenBank since 2009 had formal taxonomic names. Page termed these "dark taxa" records, the parallel in some ways of the undigitized ("dark") texts that contain species descriptions. In referring to this finding, Parr et al., (2012) [28] noted "Perhaps nothing highlights the emerging disconnection between taxonomy and genomics more than the increasing number of unnamed sequences in GenBank."[10]

The data summaries that follow divide the DNA barcode records in several public datasets into the following categories based on the fitness for use of taxonomic names:

1. Records with formal binomial species that can be connected to the published traits that distinguish members of that species from others.
2. Records with "qualified" and "provisional" binomials that aren't formal species names. These names have been curated to the degree that they can be associated with other specimens based on shared traits, but have not yet been formally described and named as a distinct species.
3. Records with a taxonomic placeholder that is not binomial in form, and may or may not be associated with other specimens.
4. Records without any taxonomic identifier.

The term "dark taxa" includes categories 2, 3, and 4. As described above, category 1 names are available in cases where frozen tissue from a well-identified voucher specimen can be obtained. Without a frozen tissue sample a collection would have to permit some destructive sampling of older tissue from the voucher, for which more advanced and complicated sequencing would be needed. If successful, the barcode record would have a category 1 name. If a new voucher had to be collected, the resulting barcode record would start with a category 3 or 4 name. As research on the specimen proceeded (aided by the barcode sequence), the name could graduate to level 2 and eventually 1. This research and data curation could be done by researchers in that project, or they could enlist help from that taxonomic community through data sharing.

In one of the examples that follow (the BARCODE 500K project), data are reported from the same dataset at two points in time after sampling has been completed. The number of records is

[10] See [27], page 96.

approximately the same, so numbers and the proportions of records (or more precisely, the taxonomic names in those records) in categories 1 and 2 provide the same information. The other examples are based on two different datasets (BOLD and GenBank) at the same point in time (June 2023). The numbers of records differ greatly between the two datasets, so proportions of names per category are more useful. For simplicity, we report data from only categories 1 (formal names) and 2 (qualified and provisional names). The numbers and proportions in categories 3 and 4 are not reported, but together they represent all the names not in categories 1 or 2.

Our goal in presenting these data is to examine the degree to which names in a database have progressed from lower to higher categories. Adding samples from unidentified vouchers will increase the size of the database but will also decrease the proportion of category 1 and 2 names. In contrast, increases in the proportion of category 1 and 2 names indicates that names are being curated (i.e., upgraded) faster than category 3 and 4 names are being added. Stated another way, changes in these proportions reflect both the volume and speed of traffic on the on-ramp to formal binomials discussed in [24].

Like the precision and usefulness of taxonomic identifications discussed above, voucher specimen identifiers can be separated into the following categories based on their fitness for use:

1. Records with voucherIDs that conform to one of several well-established standards that uniquely identifies each specimen. As a result, the barcode record points to a unique specimen record in a biorepository database.
2. Records with voucherIDs that have some, but not all, of the elements of a standard specimen identifier. It's only by interpreting the identifier that inquiries can be made at candidate repositories that might house the voucher.
3. Records with voucherIDs that bear no discernible similarities to standard specimen identifiers, so locating the true voucher is impossible without additional information.
4. Records that have no voucherIDs.

We examined these two critical elements of all barcode records (species identification and the voucher specimen identifier) for two reasons. Taxonomic assignments that are qualified by "aff" or "cf," or have provisional species names, may not be fit for either supply- or demand-side use. Voucher specimens offer opportunities to corroborate each species identification through morphological inspection and sequencing additional DNA extracts. Without unique identifiers for voucher specimens, any barcode record will be open to challenges.

All the data presented here are COI records. We have not attempted to study the standard barcode regions for plants or

Table 1
Species identifications in four barcode datasets

Data Source	Total # records	Records with formal binomials (Category 1)	Records with qualified or provisional binomials (Categories 2 + 3)	Records with no binomials (category 4)
A. Original 2016 BARCODE 500K data release	2,789,906	375,588 (14%)	40,163 (1.4%)	2,374,155 (85%)
B. 2022 BARCODE 500K data release	2,784,901	1,018,565 (37%)	178,805 (6%)	1,587,531 (57%)

A. Initial data release of the BARCODE 500K dataset on BOLD in early 2016 (http://bins.boldsystems.org/index.php/datarelease)
B. The final data release of the same BARCODE 500K dataset in 2022 following a period of data curation (http://www.boldsystems.org/index.php/datapackage?id=iBOLD.31-Dec-2016, https://doi.org/10.5883/DP-iBOLD.31-Dec-2016)

fungi. As a result, we cannot compare the progress on data curation among animal, plant, and fungal barcoding communities.

One of us (RDMP) has analyzed the curation of taxonomic names in BOLD and GenBank. Table 1 presents comparable data from two sequential views of the same data in BOLD: 2016 and 2022 versions of the BARCODE 500K dataset. For each dataset, we present the number and percent of "formal" binomials (Category 1) in each dataset that comply with the format *Genus species*. Fewer than half the barcode records have associated binomial species identifications; we have not confirmed that these are documented in authority resources such as Catalogue of Life[11] [29]. Table 1 also presents the numbers and proportions of records in each dataset that we considered "qualified" binomials (i.e., Category 2, the species names had the qualifiers "aff." or "cf."); "provisional" binomials (i.e., Category 3, a genus name followed by "sp." or "BIN" or "BOLD" or another label, followed by a combination of letter and number identifiers); and Category 4 records that had either no entry in the taxonomic name field or an uninterpretable label made of alphanumeric characters that were not in a binomial form. Category 4 records may have values in another field that places them in a higher taxonomic category, like Lepidoptera or Chondrichthyes.

Figure 1 presents the data in Table 1 in graphical form. Nearly all of the 2016 records either continue in the same category until 2022 or move to a higher category as shown in the figure. Only a few records are "downgraded" to a lower category between 2016 and 2022, presumably because further study called the original identification into question. This figure illustrates taxonomic identifications that are traveling on the on-ramp to formal taxonomic

[11] https://www.catalogueoflife.org

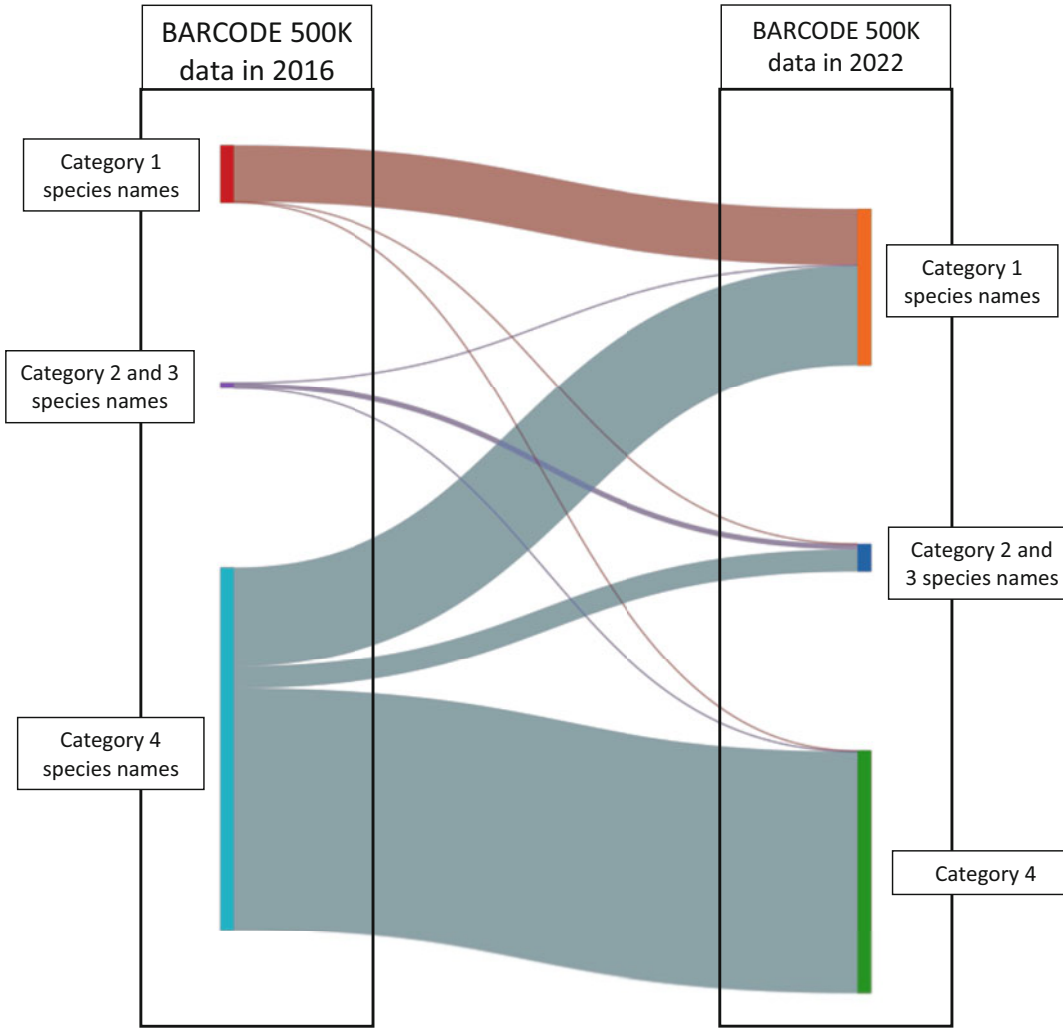


Fig. 1 Graphical representation of the 2016 and 2022 taxonomic names in the BARCODE 500K data set; see Table 1

status through taxonomic research and data curation, as described in [24].

Table 2 presents the data for two different databases (BOLD and GenBank) in the same month (June 2023). Though much larger than the BARCODE 500K dataset, the June 2023 BOLD data package showed very similar proportions of formal binomials and qualified or provisional binomials. Slightly more than half the records in the 2023 BOLD dataset had some type of binomials, while the BARCODE 500K dataset had slightly less than half. Guelph's approach to data curation appears to produce consistent results at this level.

GenBank contains 3.6 million COI records; of these 1.1 million have the reserved NCBI keyword "BARCODE," reflecting a

Table 2
Species identifications from two different databases at the same point in time (June 2023)

	Total # records	Records with formal binomials (Category 1)	Records with qualified or provisional binomials (Categories 2 + 3)	Records with no binomials (Category 4)
A. BOLD data package released June 2023	9,085,652	4,013,955 (44%)	652,267 (7%)	4,419,430 (49%)
B. BARCODE records in GenBank with BARCODE keyword as of June 2023	1,177,549	674,010 (57%)	503,539 (43%)	0 (0%)

A. The most recent data release of public records in BOLD, June 16, 2023 (https://www.boldsystems.org/index.php/datapackage?id=BOLD_Public.16-Jun-2023)
B. Download of all GenBank records with the BARCODE keyword as of June 21, 2023 (https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=nucleotide&retmax=2000000&term=BARCODE%5BKeyword%5D)

selection process based on CBOL's data standard. Implementation of the standard and application of the keyword have not been consistent, but the following analysis shows some differences from the preceding results.

The GenBank data described here include only records with the keyword BARCODE. All the GenBank records fall into Categories 1 (57%) or Categories 2 or 3 (43%). The presence of a species name is a requirement of the BARCODE data standard, which is why there are no Category 4 records among the BARCODE records in GenBank. The BOLD dataset has a lower proportion of records in Category 1 (44%) and Categories 2 and 3 (7%), and 49% in Category 4. This comparison reflects the selective application of the BARCODE keyword and the different state of formal species names and data curation between BOLD and GenBank.

DES analyzed the data curation of voucher specimen identifiers in the same datasets (see Table 3), except for the 2016 BARCODE 500K dataset; only 1% of the voucher identifiers were edited during the 2016-2022 curation period, so the 2016 and 2022 are nearly identical. Less than 2% of the records in the BARCODE 500K datasets contained at least one colon (:), the standard delimiter in the Darwin Core Triplet (DwC), and we found no "doi:" voucher identifiers. DWC [30] and DOI [31] are two systems of unique identifiers that have been discussed as a community standard. Voucher identifiers in the BARCODE 500K and June 2023 BOLD datasets are dominated by the format "BIOUG*****-E##", where * represents letters and # represents numerals. "BIOUG" is registered in GRSciColl[12] as the institution "Center

[12] https://biodiversitygenomics.net/

Table 3
Voucher specimen identifiers in three barcode datasets (sources as for Table 1)

A. From BARCODE 500K 2022 data package			B. From BOLD June 16, 2023, data package			C. GenBank BARCODE keyword records, June 2023		
Total number of records:		2,784,901	Total number of records:		8,295,166	Total number of records:		1,177,549
Number of records where museumid: Had been changed since 2016	29,646	% records 1%	Number of records where museumid:		% records	Number of records where museumid:		% records:
Had at least one hyphen	2,037,890	73%	Has at least one hyphen	5,442,371	66%	Has at least one hyphen	828,109	70%
Had at least one colon	16,187	1%	Has at least one colon	131,869	2%	Has at least one colon	287,749	24%
Includes "BIOUG"	1,669,213	60%	Includes "BIOUG"	4,622,020	56%	Includes "BIOUG"	399,281	34%
Had "SRNP"[a]	133,143	5%	Has "SRNP"	256,915	3%	Has "SRNP"	132,501	11%
Had at least one period	46,514	2%	Has at least one period	130,461	2%	Has at least one period	45,056	4%
Had "doi"	0	0%	Has "doi"	0	0%	Has "doi"	0	0%
Was blank	341,602	12%	Is blank	1,531,339	18%	Is blank	957	0%

A. Data from the 2022 data release from the BARCODE 500K project, reflecting data curation that started in 2016
B. Data from the most recent data package released by BOLD, dated June 16, 2023
C. Data from GenBank in June 2023 for all animal records with the reserved keyword "BARCODE"
[a]Dan Janzen's Costa Rican project in Santa Rosa National Park, possibly the most productive non-Guelph barcode project

for Biodiversity Genomics" at the University of Guelph, the host institution for BOLD. BIOUG's voucher identifiers don't follow the Darwin Core Triplet format because it doesn't have the traditional taxonomic departmental structure of typical specimen repositories. Instead, BIOUG specimen data are linked to barcode sequence data in BOLD, commonly assigned to BINs and higher taxonomic categories rather than traditional species names.

In contrast, approximately 70% of the voucher identifiers in GenBank's BARCODE keyword records have one or more colons, and another third use the BIOUG format. The records with one or more colons are in the Darwin Core Triplet format with institution codes registered in GRSciColl. Almost all the remaining identifiers begin with string of letters followed by number strings, some with and some without delimiters. Many of the leading letter strings are institution codes in GRSciColl and a few others are recognizable initials of researchers (e.g., "DHJ" for Dan Janzen). Access to specimen records using these identifiers in GenBank should be straightforward for institutions, collections, and individuals that are registered in GRSciColl.

5 How to Create and Feed Virtuous Cycles

Twenty years of DNA barcoding, summarized in general terms above, included some essential elements of a successful system of virtuous cycles. The following sections will highlight successful components and suggest ways to overcome impediments to broadening their success in three areas:

- Access to samples for first-time barcoding of a taxonomic group in a geographic area. These "sentinel" barcode records begin the process of accreting additional samples and attracting the interest of taxonomic specialists. These can be freshly collected specimens or existing vouchers in biorepositories.
- Data curation by taxonomic specialists who must study and integrate morphological characters of voucher specimens in biorepositories with and without barcode records. This access enables the process of refining species-level concepts and their links to barcode sequences.
- Attracting and engaging "demand-side" activities outside systematic biology that rely on dependable species-level identifications, especially in understudied taxonomic groups.

6 Virtuous Cycles for First-Time Barcode Records

One of the many barcoding initiatives led from the University of Guelph produced a data release of 1.5 million barcode records representing approximately 65,000 taxa [32]. These data came from recent large-scale collecting efforts across Canada, coupled with the high throughput sequencing facilities and BOLD informatics platform at Guelph. This confluence of researchers, lab and IT infrastructure, and funding is rare, but it provides an example of synergy among three major components that may, but don't have to, reside in one place. A significant portion of the barcode records do not have species-level identifications; BINs have been assigned with links to taxa at different taxonomic levels according to similarity to barcode records that are well identified. Further taxonomic study and data curation are needed to refine these identifications to the species level.

A different data release on GenBank [18] included 2808 bird BARCODE records from existing frozen tissue samples in a biorepository. More than 1400 of these were the first public barcode sequences for a species. Each sample was well documented, and only a few discrepancies between species identifications and barcode clusters were found. These cases will lead to further study and data curation. This demonstrates the potential of existing natural history collections for establishing new sentinel barcode records and for supporting future data curation of unidentified samples and discrepant identifications.

Thes two case studies point to the importance of institutional policies and priorities. Institutional leadership was the key ingredient for providing access, creating infrastructure, and establishing incentives for individual researchers. They also point to the importance of identifying and facilitating voucher specimens that need additional study before a formal species name can be attached to them.

7 Virtuous Cycles to Promote Access to Vouchers and Tissue Samples

Access to well-documented study specimens and tissue samples is the first step in creating reliable, verifiable reference barcode records, and despite the valuable resources in museums, herbaria, zoos, and other biorepositories, it has emerged as a rate-limiting factor for demand-side barcoding. Why are these repositories resistant to requests for access?

A US federal working group recently identified standard categories of the costs associated with the typical services that scientific collections offer to users, and connected these costs to the potential benefits generated by the use of a collection [33]. For example,

specimen digitization improves access to the collection by users, and data curation makes specimens more discoverable by potential users by exposing more and varied information about each specimen available online. A common reaction might be that digitization and data curation are costly, and increased usership may represent increased costs and staff workload with no clear compensating benefits. Some collections charge user fees to recover some of those costs, which can also reduce usership, which creates a vicious cycle.

One of the five approaches, "Value Added by Users" [33], has created a virtuous cycle that is increasing benefits generated by the collection. To do so, the collection that pioneered this approach had to increase its investments in digitization and data curation, which then increased usership.[13] In this approach, collections and institutions replace user access fees with a requirement that copies of all research products based on specimens and samples provided to users must be returned to the collection within a specific period of time. Research products include raw measurements, submissions to public datasets, digitized images, publications, subproducts such as DNA extracts or thin sections. These products are then incorporated into the collection's knowledge base, which expands the discoverability of the collection's contents to wider audiences and new uses, which increases the breadth of the collection's impact. This approach reimagines extramural users as resources for collections, rather than drains on their resources.

A recent study [34] illustrated the value that could be added to barcoding samples and records by an unanticipated source—even one with no prior connection to barcoding. A search of NCBI for a bacterial sequence found in a parasite generated a match with a mosquito COI record in BOLD. This unintended amplification led to further searches of BOLD for bacterial occurrences, and a more general rethinking of potential sources of distributional data for parasites. This study also illustrates potential innovative "off-label" uses for biomedical research of specimens and samples in natural history and other biorepositories, as proposed in [35]. All these benefits were made possible by institutional policies concerning public access and use of data.

8 Virtuous Cycles to Accelerate Data Curation

Barcode records include information contributed by field collectors, repository managers, sequencing labs, and informaticians, and these contributors are often located in different institutions and countries. Page [36] explored the value, impact, and challenges

[13] See "Value Added by Users" in [31], pages 20–23.

associated with linking diverse data types to create an integrated knowledge network for biodiversity research. Despite the challenges (or perhaps because of them), the curation process must involve these different participants, institutions, and perspectives.

Data curation, as used here, is the process of upgrading preliminary barcode data into records that are fit for use as authoritative, reliable barcode reference records. Whether used in academic research or for regulatory and legal enforcement, barcode records used for species identification will need to withstand close scrutiny. The data from BOLD and GenBank suggest strongly that there are significant numbers of barcode records that may not withstand this scrutiny without additional study and data curation. Increasing the accuracy and reliability of barcode records will require access by specialists who can then revise their taxonomic assignments and curate their public data. Can accelerate further analysis and data curation.

Virtuous cycles are based on incentives that provide benefits and/or reduce costs for the different stakeholders involved in an enterprise. What are the potential benefits of having a public information system that makes it possible to connect different pieces of information and the voucher material from which the information was extracted? Using the two case studies discussed in the previous section, two categories of benefits emerge:

- Providers (in this case, specimen collectors, biorepository institutions, collection managers) want to maximize the use of voucher specimens and their associated data, if doing so will improve the professional standing of the individuals responsible for them and the profile, esteem, and ultimately the financial support for the institution that hosts them.
- Users (e.g., downstream researchers, demand-side users, and publishers) want to maximize the impacts (academic, financial, societal, among others) while minimizing the cost and effort of finding and using curated specimens and their associated information.

GenBank and BOLD have followed different and complementary strategies in promoting growth of the DBRL. GenBank restricts the use of the BARCODE label to highlight highly curated data records. BOLD hosts barcode records at all stages of data curation and provides users with analytical tools for clustering sequences, an important aid in refining taxon concepts. To date, these resources have supported significant progress, but not an accelerating virtuous cycle. Who, then, would actually do the necessary data curation of barcode records with inadequate taxonomic identifications, and where would the improved data be hosted?

Data curation adds value to information resources like BOLD and GenBank, as well as the biorepositories that also host the

specimen records that are curated. These added values can come from researchers anywhere, whether or not they were involved in collecting the vouchers or analyzing them initially. The rate and volume of these added values depend on the institutional policies and investments in making vouchers and their associated information accessible, discoverable, and open to suggested modifications. For example, GRSciColl offers one-stop browsing for the guidelines and procedures for accessing voucher specimens and samples (e.g., loan and visiting researcher programs). By not making these guidelines and procedures available, biorepositories restrict the potential for data curation and adding value.

One of us (RDMP) points to Bionomia[14] as a model. It's an informal consortium of communities of practice (e.g., a department in an institution, or a distributed set of researchers with a common interest), more than 250 of whom have used the informatics platform created by David Shorthouse to link specimens in natural history collections (more than 21 million to date) to the individuals who have contributed to the value of those specimens (collectors, identifiers, researchers who have published on them, technicians, curators). The system uses preexisting registration systems for the specimens (like GBIF) and individuals (like ORCID) to reduce the effort needed for "scribes" to digitize the linkages. The resulting information web creates new and streamlined ways to investigate voucher specimens and their related knowledge bases, which in turn creates research opportunities.

9 Virtuous Cycles to Expand Engagement and Coverage

The barcoding facilities at the University of Guelph, BOLD, and the talented and dedicated staff that Paul Hebert has assembled have played the central and invaluable role in the growth of DNA barcoding. Under Hebert's leadership, Guelph has not only been the most productive generator of barcode records, but it has also developed a reputation as a one-stop shop for specimen curation and digitization for specimens and samples provided by non-Guelph researchers. The centralized Guelph model may have also led to the uneven representation of barcode data among countries and taxonomic groups described in [23] and [24]. As of this writing, BOLD's Taxonomy Browser states that it contains 17.4 million specimen records, of which 14.9 million are arthropods; of the 2.5 million COI records in GenBank submitted from BOLD, 2.1 million were derived from Canadian specimens. A more decentralized model may expand participation and improve coverage, and the costs and benefits are worth considering.

[14] https://bionomia.net/

Ten years of Sloan Foundation funding for CBOL ended in 2013, and new funding for a Smithsonian-based initiative was obtained through a Google Global Impact Award. The Barcode of Wildlife Project (BWP)[15] was conceived as a demonstration project for a different, more distributed approach. It focused on the use of DNA barcodes for the detection and prosecution of illegal hunting and trafficking of endangered species, but with a business plan different from Guelph's. BWP was configured to engage both supply- and demand-side stakeholders at all stages, and it used a highly decentralized approach to voucher specimens, tissue samples, lab facilities, and informatics platforms.

Potential partners in a variety of countries were approached, with the requirement that their leading biorepositories, research institutions, molecular biology and informatics facilities, and wildlife protection agencies agreed to participate. Activities were launched in Mexico, Kenya, South Africa, Nepal, Nigeria, and Ecuador beginning with multi-day workshops at which the different stakeholders shared information about their capabilities and challenges. These workshops revealed the highly varied capacity of biorepositories, molecular lab facilities, and informatics platforms among potential partner countries. They also led to the realization that enforcement of international agreements and national laws protecting endangered species varied greatly. Some countries could accept standard supply-side barcode records developed in academic research institutions, while others required the collection, management, and analysis of both reference barcodes and confiscated samples suspected to be evidence of a crime. Some countries were open to the methods for producing barcode reference records that are commonly used in academic research. Some others made no distinction between barcode records intended as reference standards and barcode based on confiscated samples. In these cases, both the supply- and demand-side barcode data had to be treated as part of a crime investigation. Law enforcement officers would have to be present from collection to final analysis to protect the chain of custody of both reference and confiscated samples, and sample storage and processing would have to be in secured law enforcement facilities. This variation in protocols required for enforcement parallels the responses that CBOL met when approaching US government agencies about possible adoption of barcodes for regulatory and forensic uses.

BWP had two ground-rules for participants: (1) no specimens or tissue samples would leave the participant countries for curation, storage, or lab analysis and (2) all barcode reference records would be submitted to GenBank with the information required for the BARCODE keyword. These records would then migrate to

[15] https://barcodeofwildlife.org/

BOLD. BWP stressed that the benefits to each country (wildlife conservation) are not in conflict with production of public barcode records that would increase the geographic and taxonomic coverage BDRL, thereby increasing its international value and usership. Releasing reference records on GenBank would also benefit the global detection, investigation, and prosecution of poachers and traffickers of species from the participant countries.

The countries in which the stakeholders demonstrated openness to implementing a barcode-based approach for detection and prosecution received training and ongoing support for lab procedures and data management; guidance as they developed country-specific protocols for collecting and curating specimens; support in the use of online informatics platforms; and assistance with fundraising.

CBOL encountered one source of resistance to engagement in DNA barcoding that seems immune to this approach. Research zoos and some other biorepositories have understandable concerns about making collection information public due to potential protests by animal rights activists, or restrictions placed on samples by donors. Years of CBOL effort went into trying to convince these biorepositories to participate in supply-side barcoding, with absolutely no success. The understandable resistance to data sharing practiced by these collections ranks alongside those practiced by classified pathogen collections.

BWP was active until 2016 but it catalyzed collaboration and projects that continue within several countries. Partnerships between supply- and demand-side stakeholders have continued, and facilities and protocols have been modernized. Some are completely within law enforcement agencies; others are hybrids between academic and enforcement institutions. We learned that each country had its own procedures and legal standards for the identification of endangered species in their trafficked forms. In these countries, the production of reference barcode records as well as the analysis of confiscated unknowns had to be conducted using chain of custody procedures and conducted in labs run by law enforcement agencies. Voucher specimens couldn't be deposited in universities or museums which lacked the necessary security measures.

The distributed model has many costs that centralized models can avoid. The lack of capacity in most developing countries requires long-term infusion of resources, both financial and technical. Lack of trust among academic institutions and government agencies are obstacles to engagement between supply- and demand-side stakeholders. Accusations of biopiracy by outsiders, and demands for benefit-sharing, can be genuine, but they can also be used as tactics for political gain by ruling and opposition parties in developing countries. On the other hand, there can be long-term

scientific, societal, educational, and diplomatic benefits for both sides if such north-south partnerships become established.

10 Some Final Observations

This chapter began by claiming that the barcoding initiative could help pave the way for whole-genome initiatives. Two examples underline this potential. The first speaks to the ongoing challenges faced by large, distributed projects that rely on long-term partnerships. The second speaks to the critical role that sharing preliminary results can play. Doing so can increase user involvement and accelerate data curation and impact, if data producers, data users, and publishers treat each other fairly.

First example: The Earth BioGenome Project (EBP) is "a moonshot for biology that aims to sequence, catalog, and characterize the genomes of all of Earth's eukaryotic biodiversity over a period of 10 years" (Lewin et al., 2018) [37]. The authors realized that "an open digital repository of genomic information for life on Earth can be realized only by a coordinated international effort," and they proceeded to outline its major challenges. Three years later, Hotaling, Kelley, and Frandsen (2021) [38] took stock of progress and reported that (1) while genome data quality had improved greatly, annotation of sequences lagged far behind; (2) the desired international effort was dominated by northern developed countries; and (3) taxonomic representation of genomes was skewed toward vertebrates and away from arthropods (in contrast to the barcoding initiative.) Among their recommendations going forward, they stated "Genome science needs specimen vouchers," a foundational principle of DNA barcoding. Their explanation included a call for data curation, similar to the one we make here:

> Vouchers serve as a key physical link between taxonomy and molecular insight. Rarely, however, are vouchers referenced in publications of genome assemblies; only 11% of vertebrate assemblies included such a reference as of January 2020. While vouchers represent a physical reference for assessing taxonomic classification or morphological variation, a properly stored voucher could also provide a long-term source of material for future resource improvement. If a physical specimen cannot be deposited, photographs and/or genomic DNA should be deposited in its place. Tied to the metadata discussion above, additional fields should be added to GenBank genome assembly accessions to directly link the assembly to a specimen, photo, or genomic DNA that has been deposited elsewhere.[16]

Second example: In January 2003, the Wellcome Trust assembled 40 representatives of the prominent stakeholders in the emerging field of genomics, including producers and users of

[16] https://doi.org/10.1073/pnas.2109019118, page 7

genomic data and public data repositories of those data. The report of that meeting (Wellcome Trust, 2003) [19] confronted the conflicting incentives that were slowing access to completed genomes. On the one hand, genomic researchers wanted to restrict access to their data until it was processed, refined, interpreted, and published. Consumers of genome data, including other researchers, wanted rapid access to the raw data so that they could refine, interpret, and publish the results. How could the global good of accelerating publication be achieved while recognizing and rewarding the contributions of the genome centers that produced the raw data? The report recommended a set of new behavioral norms for producers, funders, publishers, and users that could ensure early release of unfinished data (termed "community resources") while preserving exclusive rights for their producers to publish complete analyses for specific topics that are demarcated with the public data release.

Initially, the concept behind the DNA barcode was both mind-boggling and seemed too good to be true. Much time and effort has been devoted to the question of whether or not it's true. After 20 years of barcoding, the challenge has shifted to realizing the technological opportunities and potential for tangible, practical uses for society it offers.

Success of ambitious enterprises like DNA barcoding, the Earth BioGenome Project, and evolutionary informatics depends on the willingness of their leaders and participants to spend time outside their normal comfort zones. It requires science diplomacy and a willingness to appreciate perspectives and cultural norms of other disciplines and countries—even the divergent traditions of researchers on different taxonomic groups. It can feel like a cost, even a penalty, for pursuing ambitious goals. If patience and a broader perspective are costs, they might be the price of long-term, durable relationships that produce ambitious results, economies of scale, and decreasing costs over time.

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